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**Task Description:**

Our joint project between the University of California, Irvine and Imperial College, London explores the use of electric fields in combustion for the purpose of improving the performance of energy conversion systems. The utility of electric fields on flames can be divided in two discrete categories: sensing and actuating. As an actuator, relatively high electric fields have been shown to affect flame stability and soot formation; while as a sensing device, the electrical response, even at small electric fields, can detect the presence of a flame as well as provide characteristic information about the combustion process. In the past year, we have continued electric field testing experimentally, with special focus on a coflowing burner, both, in zero and 1g, and numerically, modeling flame structure and the influence of an electric field. These studies directly support the tests proposed for the ISS experiment. In addition, fundamental studies involving partially-admixed and quenched jet flames are directed toward understanding air quality issues involving domestic boilers and stoves. Specific activities over the year included:
1.) Irvine/NASA Glenn testing - Measuring the flame's electrical response and voltage-current characteristic under various conditions, and reduced gravity testing at the NASA Glenn 2.2-s Drop Tower.

2.) Working to develop a new sensor concept aimed at detecting the onset of quenching and pollutant release in domestic boilers by measuring the contours of first detectable ion current, which when interrupted can warn of incomplete combustion.

Task Description:

The overall goal of this project is to understand the behavior of chemi-ionization in flames and to demonstrate the use of this understanding to manipulate combustion in beneficial ways (including enhancing flame stability and reducing flame soot). The project uses detailed measurements of electrical properties of flames in several ways.

First, we can create a neutral wind (ion driven convection) locally at the flame front; this is one of the rare and important action-at-a-distance effects on flames that can be used as a localized actuator. The effect of the wind has been demonstrated with a candle flame in parabolic flight based microgravity and in a spherical cage drop tower, but there has been no systematic study of the ion wind effect in the absence of the confounding effects of the approximately equal influence of buoyancy.

Second, we can use electric fields to affect soot formation and transport – ions and soot are closely linked because there are shared chemical pathways and because soot particles are charged by the flame ions. It is possible, therefore, to modify soot formation and to use charges attached to soot to affect soot transport. By adjusting the soot trajectory through the reaction zone it is possible to affect the oxidation of soot to prevent its eventual release or to modify the residence time of soot in different zones of the flame. It is also possible to use the neutral ion wind to change the mixing behavior of the flame and reduce soot in this way.

Finally, we can use the electrical properties to characterize the flame – there is a quantifiable relationship between the carbon content of a fuel and the ion current it can produce at saturation. This relationship is already exploited in a typical flame ionization detector (FID) analyzer for hydrocarbon emission where the carbon is burned in a hydrogen flame to create an ion current in proportion to the carbon influx. In direct flame interrogation, however, there are several confounding variables that can affect the ion current, including flame temperature (which generally affects the O atom concentration) and the reaction zone volume. Flame temperature can in turn be affected by levels of premixing, diffusion rates, fuel composition, and radiative loss. It is possible, therefore, to determine flame character from the ionization curve response.

Task Significance:

Understanding electric field interactions with flames is central to combustion control, particularly in situations near limit operation of the flame where small effects are amplified dramatically. Electric fields have been shown to modify burning rates as well as to enhance and reduce flame propagation even to the point of flame extinguishment. Our experiments provide a unique ability to make simultaneous measurements of both physical and electrical properties of flames.

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

Several of the applications we explore have potential uses in earth gravity. Understanding fundamental electrical properties of flames have contributions to earth-based combustion, particularly as regards soot formation and control. Ion current is also a very useful diagnostic tool and sensor for combustion, capable of reflecting flame character and behavior for next generation power generation systems such as homogeneous charge compression ignition engines. There is also the potential to electrically sense and correct off-design combustion that releases toxic carbon monoxide during quenching. Finally, there are important potential applications in combustion synthesis of nanoparticles where ion formation pathways are important.

Experimental Testing/Development:
UCI Coflow Burner Setup

Experimental work was completed using a coflow burner. The burner consists of two stainless steel plenums threaded together. Fuel is mixed, if desired, with nitrogen in the bottom plenum and passed through a stainless steel tube (2.13 mm ID). The flow at the exit is fully developed. Air is introduced in the top chamber, passing through beads and a honeycomb mesh to create a uniform flow at the exit. A variable high potential supply creates an electric field between the burner and a mesh electrode downstream. Field strength between the electrodes is not linear due to the flame's space charge; however, the applied field strength can be defined approximately as $E = V/d$, where V is the applied potential and d is the distance between electrodes. This approximation is used in describing the experiments to provide a consistent reference to field strength. Ion current is computed using Ohm's law and the measured potential drop across a shunt resistor placed between the electrically isolated burner and the building ground. The experimental apparatus is housed in an acrylic chamber, which shields the flame from room currents.

NASA Glenn Coflow Burner Setup

At the NASA Glenn Research Center, the burner apparatus includes a cylindrical test chamber, flow system, and data acquisition/control system built into a NASA 2.2 sec. Drop Tower rig. The coflowing burner is placed coaxially inside the test chamber and the stainless-steel fuel tube is coaxially positioned inside a converging nozzle through which air

flows. Metering the upstream pressure of a critical flow orifice controls the jet flow rate. A hot-wire ignites the fuel immediately after the drop is initiated. A rotary solenoid briefly positions the igniter near the burner during ignition and away from it otherwise. To provide a uniform flow, two layers of honeycomb, two stainless steel mesh screens atop the honeycomb, and two additional honeycomb layers are placed at the base of the nozzle.

UCI Irvine Coflow Burner Results

The ion current response of a flame to an electric field shows three distinct regions: subsaturation – where the ion current response is parabolic with applied voltage, a saturation region – where the ion current plateaus and remains constant with increasing electric field, and an enhanced ionization regime – where the ion current increases beyond values predicted from a diffusion flame. As a diagnostic tool, the saturation ion current has been directly related to the carbon flux in non-premixed flames (Boothman et al., 1969). When the flame is at the saturation point, ion recombination is effectively halted, and the saturation ion current can be related to the ion production rate in the flame. In addition to the ion motion that results from an electric field, there is also a neutral ion-driven wind. The wind is the phenomenon where chemi-ions accelerated by an external electric field successively collide with the closely packed neutral gas molecules, thereby transferring their acquired kinetic energy to the bulk. The process is much like dragging a screen mesh through a stationary gas, with the gas behind the screen acquiring a velocity that is lower than that of the screen. This neutral wind can modify the local flame structure and influence ion production, which have been able to explain electric field effects identified in the literature.

Recently, we have started using CH* chemiluminescence to measure changes in flame structure. This is an important measurement to consider in the context of electrical properties of flames because the primary radical-radical reactions that give rise to chemi-ions involve CH:

CH + O --> CHO(+) + e-

CH* + O --> CHO(+) + e-

The change in CH* luminosity with applied field illustrates a bi-directional coupling between these two quantities: where the electrical body force can modify the flame structure, and this change in flame structure affects ion production, which determines the strength of the body force. Exhibited clearly in μg, this relationship makes it difficult to measure ion concentrations in a diffusion flame without the influence of an electric field. By increasing the electric field, both the ion current and CH* luminosity show a gradual increase and eventual plateau in their values. When considering the similarity between the two curves and the initial chemi-ionization reactions, which involve excited state CH radicals, it is plausible that a relationship exists between both values. This relationship, between CH* luminosity and ion current, occurs at the saturation condition and is consistent over a number of conditions. The saturation condition is a characteristic of premixed flames, much like the laminar burning velocity. It is important to recognize that it is only the saturation ion current that can be considered representative of a mixture, which is consistent with the luminosity finding. After calibrating the PMT, it should be possible to determine the ratio of CH* radicals that eventually produce H3O+ ions, as well as identifying what factors influence the ratio. A calibration of the CH* emission was accomplished by Walsh et al., 2005, in determining the absolute CH* concentration (as compared to that of unexcited CH). That work discussed the calibration method but did not include any ion current measurements. Additional tests would help determine whether CH* emission, not complicated by soot radiation, can indicate ion concentration without an applied field and may help design electrodes that directly act on these regions.

Enhanced Ionization: Positive Field

Regardless of polarity, the ion current response exhibits an enhanced ionization regime during which the ion current continues to increase beyond saturation levels. Initial measured suggested that a positive electric field accelerated the main flow, which entrained additional air through the quench zone, partially premixing the flame. Numerical simulations by Yamashita et al. [2009] that show the additional increase in ion current corresponding with an increase in the local oxygen concentration near the flame sheet support this assumption. This prediction is also consistent with work published by Weinberg and Carleton [2009], which showed that the saturation ion current of diffusion flames is fairly insensitive to aeration until reaching the rich flammability limit where the ion current shows a sharp increase.

This numerical observation was explored experimentally by raising the fuel jet above the honeycomb mesh and measuring the laser-induced fluorescence from acetone vapor seeded in the coflowing air stream using a nitrogen carrier gas. Although the amount of acetone evaporated into each bubble would have exceeded the rich flammability limit for burning in air, nitrogen was used as a precaution. The additional nitrogen had a negligible influence on the coflowing gas composition (<1%) and did not influence the ion current measurement (as confirmed by direct measurement).

Acetone's relatively low thermal decomposition temperature made it difficult to visualize near the flame sheet. Because of this, and with the exception of highly lifted flames that rested on a pocket of air, measurements did not show air leakage through the quench zone of the flame sheet. Despite the inability to directly observe leakage, the images did show the influence of an accelerating ion plume and the subsequent entrainment downstream of the flame sheet. The non-monotonic response of the coflowing jet is particularly fascinating. By gradually increasing the field strength, hot products accelerating toward the downstream electrode entrain additional gas. This can be observed as the gradual decrease in fluorescence thickness with increasing height above the burner surface. However, at 5.7 kV, the PLIF response of the coflowing gas begins to widen near the base. This same voltage also corresponds to the start of the enhanced ionization regime in ion current measurements. The coincidence between the start of the enhanced ionization regime and a change in the local flow field hints to an effect at the burner base that contributes to ion production yet counteracts the effects of the ion wind.

To isolate the source of this anomalous effect, it is necessary to study ion formation mechanisms outside of hydrocarbon combustion. Additional mechanisms that are relevant to this experiment included the burner geometry and material, the applied field strength, and local temperatures. Without a flame, a flush jet and a raised jet configuration show negligible changes in ion current with electric field. Although the system appears insensitive to geometry, it must also be considered in combination with temperature. By substituting methane with hydrogen in the same two burner geometries, and because hydrogen is non-ionizing, the effect of a heat source at the same location can be tested without the contribution of combustion ions. Below the enhanced ionization regime (< 6 kV @ 3.2 cm jet to electrode separation) of a typical hydrocarbon flame, the heat source did not contribute to ion production, regardless of geometry. However, at higher applied fields with a raised jet, the hydrogen flame configuration acted as a substantial ion source. The increase coincides convincingly with the enhanced saturation region observed in hydrocarbon flames with an upward directed

field. The source of these ions is not clear; however, it is clearly aggravated by the heat source making thermionic emission from hot surfaces a likely major contributor to the secondary ionization region of the V-I curve. This is a complication that has not been identified in the past and has important implications for other studies with flames near surfaces where all ion current (and the resulting ion wind) is attributed to chemi-ionization.

Enhanced Ionization: Negative Field

A similar enhanced ion current behavior is present with a negative field. The Voltage-Current Characteristic (VCC) curves for a CH4/air mixture in the 1g coflow burner for a number of electrode separations, the separation between the electrodes can modify the slope and onset of this enhanced ionization region when the field is directed toward the burner. Advanced onset of enhanced ionization with electrode spacing is likely a result of electrons traveling downstream with high energy and colliding with neutral molecules to create additional electrons and ions. To visualize this assumption, consider a particle moving through the electrode gap with an initial kinetic energy. If this particle were responsible for an ionizing collision, the maximum conversion of this initial kinetic energy to internal energy would result from an inelastic collision with a second particle.

By taking the ratio of these two energies, it is evident that the greatest conversion of kinetic to internal energy occurs when M1 << M2. A likely particle candidate for ionization by collision would be an electron. If we assume the influence of space charge on the potential field distribution is minimal, it would be expected that an electron traveling over a larger electrode gap with the same potential gradient would have more collision opportunities that would lead to secondary electrons.

By selecting an efficiency factor assuming the enhanced saturation region begins at E0 = 2.3kV/cm, the calculated ion current response agrees remarkably well with the experimental measurement indicating that a major contributor to the secondary ionization region of the V-I curve with a downward-directed electric field is from electrons colliding with neutral molecules in the unipolar region with an energy sufficient to create an ionizing event.

**Task Progress:**

Methane flame studies at NASA Glenn

In a 1g environment, the plume of hot gases from a flame naturally induces, through mass continuity, a convective flow to the neighboring cooler environment. This natural convection gives rise to a flame's egg-like shape in Earth gravity. In microgravity environments, buoyant convection, which is mainly responsible for the transport of fresh oxidizer to a flame, is removed and the flame is mainly driven by molecular diffusion and forced convection [Ross, 2001]. A number of experiments have studied the influence of an electric field in microgravity environments to measure ion mobility [Yuan and Hegde, 2003], burning rate [Yamashita et al., 2008], and the feasibility of inducing convection in the absence of gravity [Carleton and Weinberg, 1987]. This section describes new experiments involving a jet flame under the influence of an electric field in 1g and µg using the NASA 2.2-s Drop Tower to identify the role buoyancy-driven convection and ion wind-driven convection play in controlling flame shape.

By aligning an applied electric field with the main flow of the jet in a µg environment, the resulting ion wind drives away combustion products, much like buoyancy, while entraining additional air to the reaction zone. Images extracted from videos show steady flames taken under different gravity and voltage conditions, with the exception of the µg flame without an applied field. Without an electric field or forced convection, a µg flame requires a longer settling time than is available on the drop tower and the flame continues to expand over the drop duration. Without external influences, a low initial momentum µg flame takes on a near spherical shape owing to the lack of a preferential transport direction of products. This is in sharp contrast to a 1g environment where buoyancy is a dominating force and can overwhelm the electric field influence.

The relative contribution of gravity, forced convection, and electric field to flame area and location were compared. The flame was idealized as an oblate spheroid to calculate area and was compared to a 1g flame without an electric field or coflow as a reference. Images were taken from drop videos when the flame appeared stable. Images show a small impact of an electric field on flame area and location in 1g, while in microgravity there is a clear similarity between the impact of the electric field and forced convection on a flame's shape and area. Results show that a flame in µg with a 0.6 kV/cm average applied electric field or a coflow velocity of 24 cm/s will produce a flame with a similar area and flame base location as an unforced 1g flame. In addition to area calculations, similarity was also measured by taking the absolute difference between a microgravity image and a 1g reference flame and computing the RMS of the entire differenced image. When used for image comparison, the RMS metric is a quadratic mean that assumes that a reference image is known, which in this case is the 1g flame, while the µg flame is a distorted version of this reference. A low computed RMS value corresponds to strong similarity between images. Both the images and computed RMS values show the closest match to be between the microgravity flame with -5 kV (1 kV/cm) applied and a 1g flame without an applied field. Carleton et al. [1998] observed in their research that similar field strengths were necessary to simulate microgravity in an Earth based laboratory environment. Differences in soot luminosity were not evaluated because of pixel saturation; however, future work will include the use of a filter to normalize the color channels, improving comparisons. Understanding the differences between the two comparison techniques as well as the potential to infer similarity from Schlieren images will continue to be explored.

Quenching studies

The research underway at Imperial College examines the effects of quenching on flame structure and understanding the hazardous conditions created by poorly tuned industrial boilers. In our studies, appreciable concentrations of carbon monoxide appear within the flame cone, particularly for burners with a spatial scale of the order of a diffusion flame thickness, which raised concerns regarding the potential hazard that if by partially quenching the flame, dangerous levels of CO were released into the ambient atmosphere.

Initially the burner was a copper tube, with a 4 mm internal and 6 mm external diameter (since gas heaters in boilers often consist just of multiple apertures in metal tubes, thick – walled, small diameter burners were quite appropriate for the planned quenching measurements). Previous experimental results, as well as those obtained by numerical simulation, imply that, while no CO would emerge into the atmosphere if the flame reaction proceeded to completion normally, CO escape might pose a serious hazard if quenched before completion - particularly if that occurred close to the peak of CO concentration arrested the reaction. During 1999-2004, CO poisoning caused 16,447 deaths in the United States; the principal cause being incomplete combustion in malfunctioning heaters/boilers.

Current measurements were plotted against electrode separation for a flame burning 82.75 ml/min natural gas on a sharp-edged burner with a 4 mm inner diameter. The burner is one electrode, and gauzes previously used for quenching

measurements are the other. The two metals, stainless steel and aluminium, were chosen for their different equilibrium temperatures and quenching characteristics (see previous work; also Weinberg et al., Combustion and Flame 158 (2011) 273). The voltage of 38.7 V is delivered by four nominally "9 V" batteries in series. The graphs show the effect, for both polarities, of lowering the gauzes, starting from the product gases at a height well above both, the luminosity of the flame and any CO release by quenching.

While the apparatus is very simple, the electrical structure is quite complex. The reaction zone is the source of all electrical charges. The field strength, and hence the current, increases as the distance between the electrodes decreases. The local current density is j = K± E e n±

where K is the charge carrier mobility and E, e, n, represent electric field vector, fundamental charge, and number density respectively. The sign in the suffix denotes the polarities over which j is summed – (see e. g. Lawton and Weinberg (1969) "Electrical Aspects of Combustion," Clarendon Press, Oxford)

In the absence of soot particles (which we tried to avoid because of their very low mobilities) there are two types of charge carriers: electrons and ions. Electrons have a mobility at least two orders of magnitude greater than ions and hence carry most of the current. All positive charges are ions; negative charges are electrons initially, though they can attach to form negative ions, particularly if they travel through cold gas for more than a very few millimetres.

The other regions are the hot inert product gases above the reaction zone in which no charge is generated and the two narrow regions which are cooled by the burner rim and the quenching gauze. Neither of the two grids appear to emit electrons thermionically.

Starting with the gauzes at over 30 mm above the burner and lowering them to the top of the reaction zone at around 17 mm above the burner, with the burner negative, electrons traversing the hot products from the reaction zone encounter little resistance and the current rises. With the burner positive, however, the resistance to positive ions is too great for the current to flow and the system acts as a rectifier. This ceases to be the case once the gauze contacts the reaction zone. Below this position the current rises steeply and continuously.

For our first objective above, it is sufficient to note that the electrical detection of proximity to the flame of a quenching surface should be based on a positively charged burner to utilize the further electron penetration. The results show that as a "quench detector" with the burner +ve, the current signals the approach to a quenching surface at a height which is more than about 2 mm in excess of that at which any quenching is ever recorded.

In practice, of course, quenching may occur by surfaces not directly above the flame. To explore the loci of electrical quenching detection, a rod with a rounded end was used as the negative electrode. It was clamped in the x-y-z positioner and approached the flame orthogonally at locations around its periphery. The contours of the first detection of a current for two potentials (37.6 V and 60 V) lie well outside the regions in which quenching had previously been measured. These results show clearly that detecting electrically the proximity to the flame of quenching surfaces, to give a warning before any carbon monoxide is released, is feasible. It can be achieved using only small batteries and could readily be developed into a system which would give warning of hazardous conditions arising from, for example, an unexpected elongation of the flame due to an obstruction in the feed.

Numerical Simulations

Recently, we have started exploring the effect of electric fields on combustion using OpenFOAM, an open source software package. The strong coupling between the chemistry, flow field, and electric field, combined with the difficulty in directly probing the flame due to field interference makes numerical simulations a potentially valuable resource to consider. OpenFOAM (Open Field Operation and Manipulation) is a collection of partial differential solver libraries, written in C++ that can be easily adapted to a number of interesting problems, including combustion. Our model implements the use of reactingFoam, which is an explicit turbulent reacting flow solver that we have adopted for laminar simulations and have included electric fields interactions in the fluid flow. To solve this deeply coupled problem, Poisson's equation must be solved for the electric potential distribution with the influence of space charge, the Navier-Stokes equation includes an additional term to account for the electric body force on the neutral gas by the colliding ions, and the species transport equation includes the effects of ion drift velocity on the distribution of charged species. The problem is solved with a 2D axisymmetric domain modeled in a coflowing geometry with a 2.1 mm inner jet and a 25.4 mm outer jet (matching the geometry of the experimental apparatus). The downstream boundary, which also serves as the mesh electrode, is 5 cm from the base of the burner. No radiation model is implemented. Numerical simulations were performed using GRI-Mech Version 3.0 with nitrogen species neglected for the neutral mechanism and additional chemistry by Prager, et al. (2007) and Walsh, et al. (1998), which include charged species and CH* radical chemistry. Species diffusion is computed using a mixture averaged relationship proposed by Mitchell (1979). The simulation assumes a unity Schmidt number and a 0.7 Prandtl number, which is a sufficient assumption for initial simulation. Future work will include the effects of mixture-averaged transport properties. Predicted temperatures are higher than expected, which can be attributed to the lack of a radiation heat loss model. Flame heights are larger than experimentally observed heights and predictions by Roper. Nevertheless, the results predict a peak in ion formation at a reaction kernel located at the base of the flame. This location coincides with peak heat release and CH concentration, which are consistent predictors of ion production. Identifying that ion production can be localized to specific regions, sheds light on the ionization process and will aid in creating simplified models to describe ion/flame interactions. This result also lends insight in understanding why the majority of the observed effects (e.g. flame stabilization and oscillations), with the exception of highly sooting flames where flame tip deformation is observed, are localized at the flame base.

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| **Bibliography Type:** | Description: (Last Updated: 11/26/2025) |
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